Full-Dimension MIMO (FD-MIMO) for Next Generation Cellular Technology

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ABSTRACT

This article considers a practical implementation of massive MIMO systems [1]. Although the best performance can be achieved when a large number of active antennas are placed only in the horizontal domain, BS form factor limitation often makes horizontal array placement infeasible. To cope with this limitation, this article introduces full-dimension MIMO (FD-MIMO) cellular wireless communication system, where active antennas are placed in a 2D grid at BSs. For analysis of the FD-MIMO systems, a 3D spatial channel model is introduced, on which system-level simulations are conducted. The simulation results show that the proposed FD-MIMO system with 32 antenna ports achieves 2–3.6 times cell average throughput gain and 1.5–5 times cell edge throughput gain compared to the 4G LTE system of two antenna ports at the BS.

INTRODUCTION

In order to meet growing demands of wireless data, Third Generation Partnership Project (3GPP) Long Term Evolution (LTE) and LTE-Advanced Releases 8 through 11 have introduced cutting edge techniques to improve spectral efficiency, such as multiple-input multiple-output (MIMO), coordinated multipoint (CoMP) transmission/reception, and heterogeneous networks. However, to catch up with ever growing future demands, new technologies to further boost the spectral efficiency are strongly desired for cellular wireless communications.

One promising technology for meeting the future demands is massive MIMO transmission/reception [1]. When used with multi-user precoding schemes such as maximum ratio transmission (MRT) precoding (a.k.a. channel conjugate precoding) [1], massive MIMO systems experience small inter-user and intercell interference, and consequently achieve significantly higher throughput than the state-of-the-art MIMO systems.

One of the main challenges to build massive MIMO systems in practice is that the number of antennas that can be equipped at a base station (BS) is often limited by the BS form factors and operating carrier frequencies. For example, to install a large number of antenna elements (e.g., > 8) horizontally at the top of a BS tower operating with the lowest LTE carrier frequency of 700 MHz, just eight antenna elements with 0.5 λ spacing require a horizontal room of ~1.7 m, where λ is the carrier wavelength. For typical LTE carrier frequency of 2.5 GHz, fitting 32 antenna elements with 0.5 λ spacing require a horizontal room of ~1.9 m, which is still not feasible in many BS that has only limited room on the tower. This practical limitation has motivated full-dimension MIMO cellular communication systems, or FD-MIMO [14, 15], which places a large number of active antenna elements in a two-dimensional grid at the BSs, as shown in Fig. 1.

Traditionally, the wireless communications standards community; for example, 3GPP has utilized 2D spatial channel models (2D SCMs) on the horizontal cross-section of the wireless channels for evaluating the standardized technologies operating with BSs equipped with horizontally placed antennas. For evaluation and design of FD-MIMO, however, such 2D channel models are not adequate and three-dimensional spatial channel models (3D SCMs) are necessary. The 3D SCM should take into account azimuth direction as well as the elevation direction of signal propagation between a BS and a UE device. Although the 3D SCM has been investigated in WINNER projects and literature (e.g., [2–5]), to the best of the authors' knowledge, currently a complete and reliable 3D SCM is not yet available. To this end, this article introduces a 3D SCM being built based on an extensive study of existing literature, and uses the resultant model for performance evaluation of FD-MIMO.

TWO-DIMENSIONAL ACTIVE ANTENNA ARRAY FOR FD-MIMO

Recently, significant interest has risen on exploring the elevation domain as well as the azimuth domain in devising MIMO transmission/reception schemes. In particular, passive or active antennas are placed in the elevation domain to provide additional degrees of freedom, which

Part of the contents in this article were presented in [12, 14].

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are effectively generated by elevation spread of the channels and the elevation angle distribution of users. The elevation domain antennas are used to form multiple elevation domain beams. For example, these elevation domain beams can be used to form subsectors in the elevation domain across the system bandwidth, where the subsectors adapt slowly depending on the user load and locations. In another example, these elevation domain beams are frequency-selective and can adapt quickly to cope with various user traffic and conditions, which allows fully flexible multi-user MIMO exploiting the full 3D channels (i.e., FD-MIMO). Some recent deployments taking advantage of elevation domain beamforming (e.g., [8, 9]), have shown up to 30 percent gain in system capacity. However, most commercial solutions so far have considered a smaller set of antenna ports (typically up to eight) to be compatible with the current specifications of LTE; this is also due to cost and space constraints.

We envision that an antenna array with a much larger number of antennas can be designed and demonstrate the potential system gains. A 2D antenna array warrants careful consideration due to the BS form factor limitation if a practical massive MIMO system is to be built. Fully flexible digital precoding requires as many transceiver chains as the number of antennas elements. This introduces additional hardware requirements to handle the large number of transceiver chains. Furthermore, it is also important to consider the backhaul capacity as a larger number of transceiver units require a correspondingly larger backhaul capacity, which can be expensive. If the antennas are located on top of the tower, the backhaul should be able to carry the highly sampled waveform data. Standardized interfaces like the common public radio interface (CPRI [10]), which can support up to 10 Gb/s, are one option. New designs may also be needed considering the coupling between different transceiver chains and implementations of efficient routing from baseband to radio frequency (RF) front-end. However, since the power requirements of individual RF chains are now reduced, some cost benefits can also be realized. The miniaturization of electronic components due to advancement of technology in filters, digital-to-analog converters (DACs), and power amplifiers (PAs) also allows practical realization of FD-MIMO at a reasonable cost and size.

Based on the above considerations as well as the observations from performance evaluations, we propose and study a 2D antenna array configuration comprising 8 antenna ports in the horizontal dimension and 4 antenna ports in the vertical dimension, giving a total of 32 antenna ports. The horizontal antenna spacing is 0.5λ and the vertical antenna spacing is 0.2λ to reduce the width of elevation beams, as discussed later. Each antenna port consists of a four-antenna vertical sub-array to provide enhanced directional antenna gain in the elevation domain. At carrier frequency of 2.5 GHz, it is possible to construct an FD-MIMO 2D antenna array system measured at ~0.5 m horizontally and ~1 m vertically, which can comfortably fit on a macro cell tower.

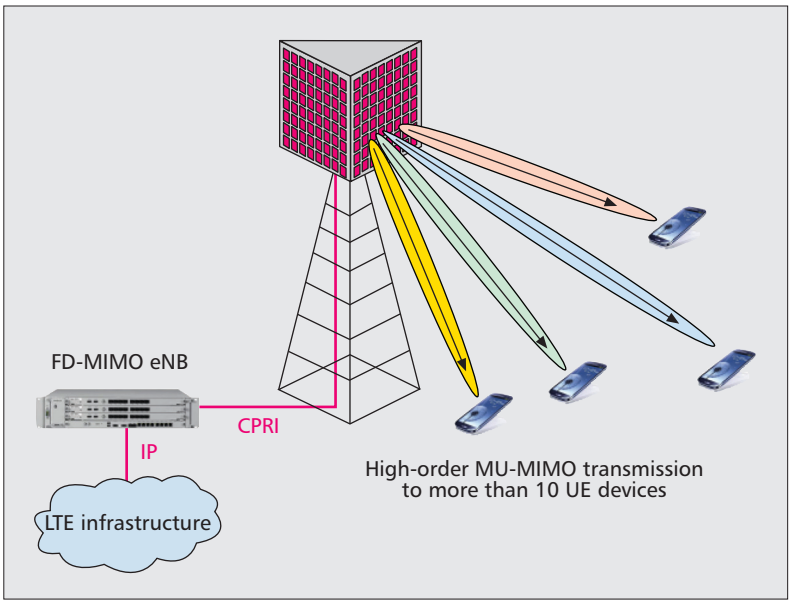


Figure 1. *The FD MIMO system.*

Three-Dimensional Spatial Channel Model

The propagation effects of a wireless channel can be modeled with a large-scale propagation model combined with a small-scale fading model. The large-scale propagation model predicts long-term slow-changing characteristics of the wireless channel, such as path loss and shadowing. The small-scale fading model provides rapid fluctuation behavior of the wireless channel due to multipath and Doppler spread. For a wireless channel with multiple antennas (MIMO channel), static beamforming gain such as the sectorization beam pattern also contributes to the long-term propagation characteristic of the wireless channel and can be modeled as part of the large-scale propagation model. As for the small-scale fading model of a MIMO channel, the correlation of signals between antenna elements also needs to be taken into account, and can be modeled by an SCM.

The SCM is a geometry-based stochastic channel model that was developed by the 3GPP-3GPP2 SCM Ad Hoc Group [6] and is widely used by the 3GPP community to perform system-level performance evaluation of MIMO technologies. Although radio beam patterns generated by BS antenna arrays are 3D in nature, antenna arrays at the BS are usually modeled as linear horizontal arrays, and elevation angles of signal paths have been ignored for simplicity. In this sense, the SCM used in the 3GPP community is a 2D model, where the elevation angle (Fig. 2b) of each signal path is always assumed to be zero (see left of Fig. 2a). The 3D SCM models non-zero elevation angles associated with signal paths as well as azimuth angles (see right of Figs. 2a and 2b) so that the small-scale fading effect on each antenna element of the 2D antenna grid as well as the correlation between any two pair of antenna elements on the 2D antenna grid can be modeled.

Although only 2D SCM has been considered

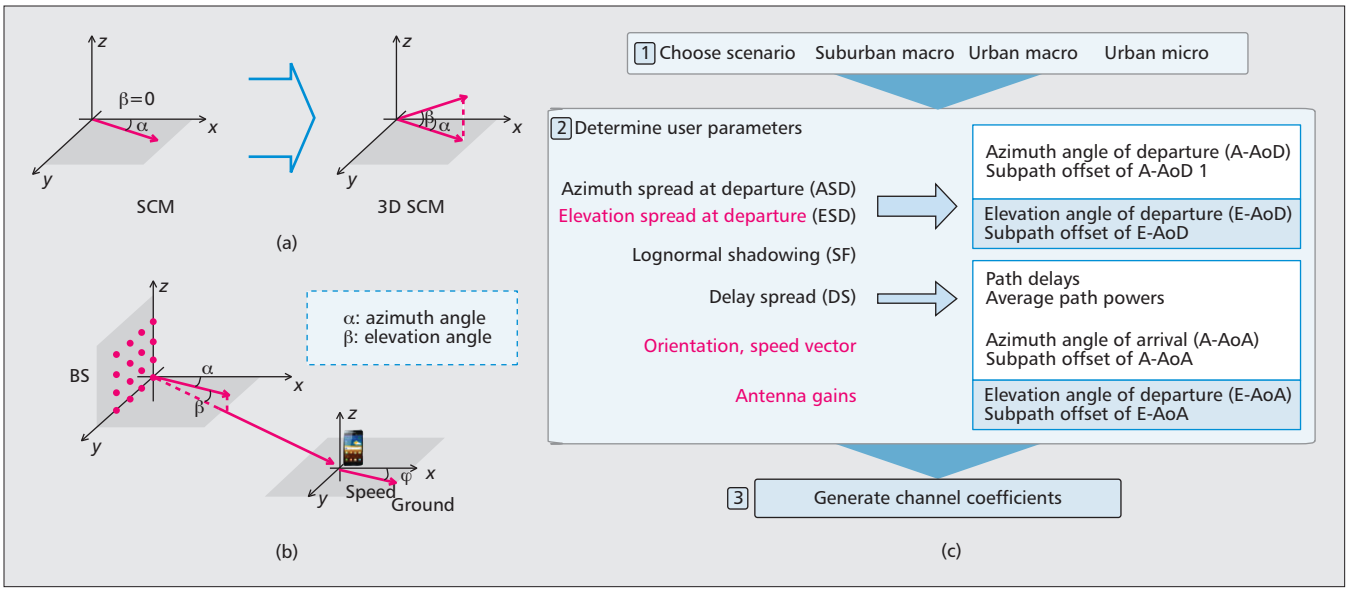


Figure 2. *Extension of 2D SCM to 3D SCM.*

in the 3GPP thus far, 3D modeling of the wireless channel has been considered in 3GPP in the form of a large-scale 3D sectorization beam pattern, which is constructed with a horizontal and a vertical antenna patterns. The 3D sectorization beam pattern has been used for determining the coverage of each cell. For a three-sector macrocell, the horizontal and vertical antenna patterns are commonly modeled with a 3 dB beam width of 70° and 10°, respectively, with a peak gain of 17 dBi for the combined antenna pattern [13]. The vertical antenna pattern is also a function of electrical and mechanical antenna downtilt, where the electrical downtilt is a result of vertical analog beamforming, produced by applying a common and static phase shifts to each vertical array of 2D antenna elements. For instance, the electrical downtilt is set as 15° and 6° for a macro deployment scenario with inter-site distance of 500 m and 1732 m, respectively, in a 3GPP evaluation methodology [13]. For a conventional BS where the vertical analog beamforming is applied after the power amplifier, the electrical downtilt is typically fixed or statically configured; this has been the case modeled in 3GPP studies so far.

For a BS equipped with columns of active antenna arrays where vertical analog beamforming is applied before the power amplifier, the electrical downtilt can be dynamically adjusted in time; this flexibility allows the BS to dynamically adjust its cell coverage depending of the traffic distribution in the cell at the time (e.g., analog beamforming can be directed toward UE devices congregated at the same location). Nonetheless, such beamforming is still fundamentally cell-specific (i.e., spatial separation of individual UE is not possible) and operate on a long timescale, albeit time varying. Therefore, dynamic analog beam patterns can simply be modeled as dynamic sectorization beam patterns, and 3D SCM may not be necessary for performance evaluation.

On the other hand, FD-MIMO involves precoding/beamforming exploiting both horizontal and vertical degrees of freedom on a small timescale in a frequency-selective manner. Hence, 3D channel modeling is required for performance evaluation of FD-MIMO. In this article, we propose to build a 3D SCM based on the same modeling principle as in the (2D) SCM proposed by 3GPP/3GPP2, additionally taking into account elevation domain parameters. An overview of the three steps to generate channel coefficients is shown in Fig. 2c and below:

Step 1: Specify an environment to be simulated, such as suburban macro, urban macro, or urban micro.

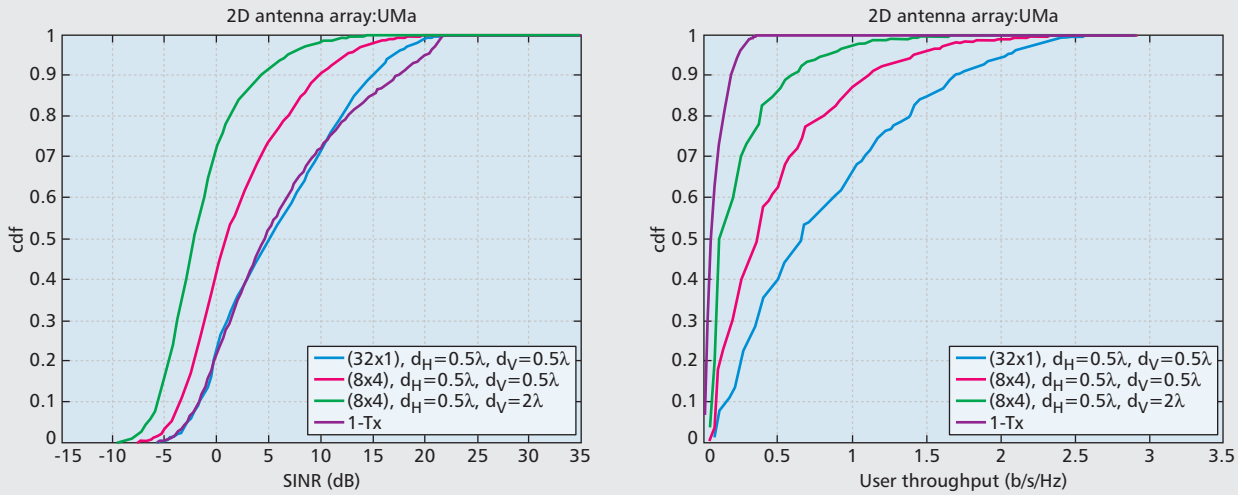
Step 2: Determine the UE placement with respect to the BS and the user parameters to be used in the simulation according to the specified environment, including azimuth and elevation spread at departure, azimuth and elevation spread at arrival, time delay spread, lognormal shadowing, user velocity, and path loss.

Step 3: Generate the channel coefficients and path delays based on the user parameters in step 2.

The channel of the nth path ($n = 1 \ldots N$) between the uth MS antenna and sth BS antenna, assuming M subpaths per path, is

$$h_{u,s,n}(t) = \sqrt{\frac{P_n \sigma_{SF}}{M}} \sum_{m=1}^{M} \left(\begin{array}{l} \sqrt{G_{BS}\left(\theta_{n,m,AoD}, \beta_{n,m,AoD}\right)} \\ \times \exp\left(j\left[kd_s \bar{r}_s \cdot \bar{\Phi}_{n,m} + \Phi_{n,m} \right] \right) \\ \times \sqrt{G_{MS}\left(\theta_{n,m,AoA}, \beta_{n,m,AoA}\right)} \\ \times \exp\left(jkd_u \sin\left(\theta_{n,m,AoA}\right) \right) \\ \times \exp\left(\begin{array}{l} jk\|\mathbf{v}\| \cos \beta_{n,m,AoA} \\ \cos\left(\theta_{n,m,AoA} - \theta_v\right) t \end{array} \right) \end{array} \right)$$

where $\bar{r}_s \cdot \bar{\Phi}_{n,m} = x_s \cos \beta_{n,m,AoD} \cos \theta_{n,m,AoD} + y_s \cos \beta_{n,m,AoD} \sin \theta_{n,m,AoD} + z_S \sin \beta_{n,m,AoD}$. The notations are as follows:



- 57 sectors, 10 UE/sector
- Long-term fading only
- Perfect AoD assumed at eNB
- SINR-throughput mapping using TU6 3km/h link results

BS antenna configuration	1Tx	(32x1), $d_H=d_V=0.5\lambda$	(8x4), $d_H=d_V=0.5\lambda$	(8x4), $d_H=0.5\lambda d_V=2\lambda$
Half power beamwidth (H, V)	(70, 10)	(90, 90)	(90, 90)	(90, 90)
Cell average throughput (bps/Hz)	0.880	8.335	2.672	5.106
Cell edge throughput (bps/Hz)	0.010	0.111	0.068	0.091

Figure 3. *1D vs. 2D antenna configurations.*

$\beta_{n,m,AoD}$: Elevation domain angle of departure (E-AoD) for the mth subpath of the nth path at the BS with respect to the BS broadside

$\beta_{n,m,AoA}$: Elevation domain angle of arrival (E-AoA) for the mth subpath of the nth path at the MS with respect to the MS broadside

$\theta_{n,m,AoD}$: Azimuth domain AoD (A-AoD) for the mth subpath of the nth path at the BS with respect to the BS broadside

$\theta_{n,m,AoA}$: Azimuth domain AoA (A-AoA) for the mth subpath of the nth path at the MS with respect to the MS broadside

$\bar{r}_s$: Location vector of Tx array element s

x_s, y_s, z_s: Components of $\bar{r}_s$ to x, y, and z-axis respectively

$\bar{\Phi}_{n,m}$: Departure angle unit vector of ray n, m

P_n The power of the nth path

s_{SF}: The lognormal shadow fading, applied as a bulk parameter to the N paths for a given drop

$G_{BS}(\cdot)$: The BS antenna gain of each array element

$G_{MS}(\cdot)$: The MS antenna gain of each array element

k: The wave number $2\pi/\lambda$ where λ is the carrier wavelength in meters

d_s: The distance in meters from BS antenna element s from the reference ($s = 1$) antenna

d_u: The distance in meters from MS antenna element u from the reference ($u = 1$) antenna

$\Phi_{n,m}$: The random phase of the mth subpath of the nth path

$\|\mathbf{v}\|$: The magnitude of the MS velocity vector

θ_v: The azimuth angle of the MS velocity vector with respect to the MS broadside

In the 3D SCM, parameters characterizing the paths are generated according to correlated spatial channel statistics: shadowing, delay spread, azimuth spreads at departure (ASD), and elevation spreads at departure (ESD). The E-AoD is generated according to a marginal distribution with the variance of ESD, and the E-AoA is generated according to the relative power of the path. Then each path is further split into several subpaths. The elevation angles of subpath n corresponding to path m, $\beta_{n,m,AoD}$ and $\beta_{n,m,AoA}$, are determined by the elevation angles of path m, the line-of-sight (LOS) angle, and a fixed cluster angle spread. For generating ESD, the 3D SCM mainly relies on elevation domain statistics reported/suggested by WINNER II, WINNER+ [2, 3], and COST273 [7].

Performance Evaluation of FD-MIMO

This section presents system-level simulation results on the performance gains of FD-MIMO over the traditional MIMO systems. The deployment scenario is according to the 3GPP case 1 [13], where 19 cell sites and 57

sectors (i.e., 3 sectors per cell site) are deployed in a hexagonal grid, each serving 10 UE devices on average. It is noted that the evaluations in this article do not take pilot contamination effects [1] into account as the number of antennas at the BS assumed for the simulations is not large enough such that pilot contamination becomes the dominating factor [11].

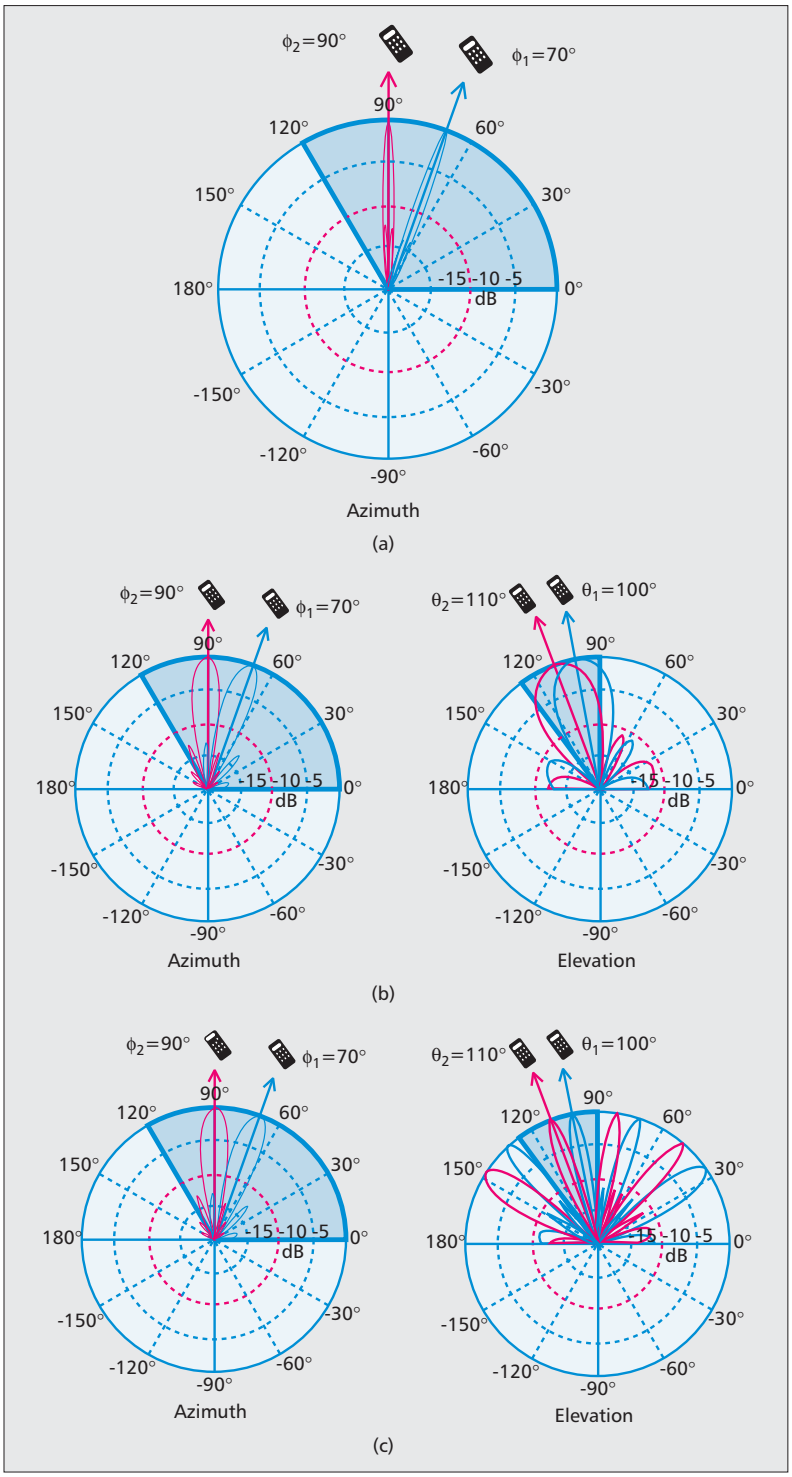


Figure 4. *FD MIMO beam patterns in azimuth and elevation domains (isotropic arrays assumed for the illustrative patterns): a) 1D (32H × 1V): d_H = 0.5 λ; b) 2D (8H × 4V): $d_H = d_V$ = 0.5 λ; c) 2D (8H × 4V): d_H = 0.5 λ, d_V = 2 λ.*

1D vs. 2D Antenna Configurations: Insights from Line-of-Sight Channels

In this section, we evaluate the system performance under line-of-sight (LOS) channel condition with only large-scale propagation model, i.e., without small-scale fading models. This type of evaluations are useful for obtaining quick insights into the new 2D antenna configuration.

Various 1D and 2D antenna configurations having a total of 32 antenna elements at each BS are considered as in Fig. 3: 32H × 1V with 0.5λ spacing, 8H × 4V with 0.5λ spacing for both horizontal and vertical dimensions, and 8H × 4V with (0.5λ, 2λ) spacing, respectively, for horizontal and vertical dimensions. Here, aH × bV implies a horizontal digital antenna ports and b vertical digital antenna ports, resulting in a total of $a \cdot b$ digital antenna ports. It is further assumed that each BS has perfect knowledge of angles of departure to the serving UE devices, and each UE device also has perfect channel state information (CSI). In order to separate the impact of the small-scale fading from the final results while at the same time capturing important characteristics of the 3D channels, only long-term large-scale channels with antenna correlations resulting from the respective antenna configurations are modeled according to International Telecommunication Union (ITU) UMa channel models [13]. In the case of FD-MIMO, each BS applies the maximal ratio transmission (MRT) precoding to each UE device, while scheduling all its serving UE devices with the full bandwidth in all scheduling instances (denoted as *all-user full-BW scheduling*). In case of the reference 1-Tx case, MRT precoding with proportional fair scheduling is utilized. The resulting signal-to-interference-plus-noise ratio (SINR) cumulative distribution function (CDF) curves for the different antenna configurations are obtained as on the top left of Fig. 3. The SINR CDF curves are converted to throughput CDF curves on the top right of Fig. 3 by utilizing an SINR-throughput mapping curve obtained from link-level simulations with typical urban 6-path (TU-6) channel models and 3 km/h UE speeds. The table at the bottom of Fig. 3 summarizes the resultant cell average throughput and cell edge (or 5-percentile) throughput.

Figure 3 shows that the SINR CDF for an FD-MIMO system with 8H × 4V and (0.5λ, 0.5λ) spacing is much worse than the 1-Tx system. This is because the all-user full-bandwidth scheduling results in the largest multi-user interference with such an antenna configuration of all the FD-MIMO antenna configurations considered in this section. On the other hand, Fig. 3 also shows that the FD-MIMO system with 8H × 4V and (0.5λ, 0.5λ) spacing achieves a better *throughput* CDF than the 1-Tx system despite the fact that the SINR CDF is worse. This series of results demonstrate that the increased scheduling opportunity (a pre-log factor in the Shannon capacity), thanks to the all-user full-bandwidth scheduling scheme, successfully overcomes the SINR loss (a post-log factor) due to the MU-MIMO interference. In this case, the increased scheduling opportunity is a 10× pre-log factor, when compared to the 1-Tx system

where a UE device can be scheduled only with one tenth of the time-frequency resources available in each cell when such a fairness constraint is imposed.

Figure 3 shows that when antenna elements are spaced apart by 0.5λ, 1D (32H × 1V) configuration achieves significantly higher throughput than 2D (8H × 4V) configuration. Figure 4 explains the reasons for such results. In the case of 1D configuration of 32H × 1V as shown in Fig. 4a multi-user interference is virtually zero as the beamwidth formed by the MRT precoding is narrow, and hence MU-MIMO with K-user scheduling achieves almost K times user throughput as the single-user MIMO (SU-MIMO), where K is the number of users in each cell. Having placed antennas in 2D (i.e., 8H × 4V), both azimuth and elevation beam patterns are necessary to be considered for understanding the performance results. Figure 4b shows the beam patterns of the 8H × 4V system with 0.5λ spacing. In the azimuth domain, even though the beam width has become wider than the corresponding 1D system of 32H × 1V, it is relatively easy to find two UE devices whose beams are sufficiently separated such that the multi-user interference is small in the sectorization angle range of [0°, 120°] (e.g., two users located at azimuth angles $\phi = 90°$ and $\phi = 70°$). However, in the elevation domain the vertical angle range is much narrower (e.g., [90°, 130°]), where 90° corresponds to the elevation domain angle of a UE device placed infinitely far away from the serving BS. Hence, it is more difficult to find two UE devices whose beams are sufficiently separated in the elevation domain; for example, when two UE devices are located at elevation angles $\theta = 100°$ and $\theta = 110°$, the multi-user interference is non-negligible.

To cope with the issues regarding the elevation domain beamforming, an alternative approach of placing vertical domain antennas at a wider spacing (e.g., 2λ vertical spacing) is considered as described in the previous section. When antenna spacing larger than 0.5λ is used, grating lobes are generated, and consequently the beamwidth becomes narrower in the elevation angle range. Figure 4c shows the beam patterns generated by the 8H × 4V system with 2λ vertical spacing, and it illustrates that the beam width of the elevation beam is much narrower than the corresponding 8H × 4V system with 0.5λ vertical spacing. The average cell throughput and the cell edge throughput achieved by the 2λ vertical spacing are shown in the table in Fig. 3, and it can be seen that the spectral efficiency numbers achieved by the 8H × 4V system with 2λ vertical spacing is significantly larger than the corresponding the 8H × 4V system with 0.5λ vertical spacing.

Figure 5 shows trends of the cell average throughput and cell edge throughput of the 8H × 4V FD-MIMO system with 2λ vertical spacing, with increasing number of UE devices per cell, or K. The average throughput increases until K reaches ~15, and decreases as K further increases. Although the cell edge throughput keeps decreasing as K increases due to the increased intra-cell multi-user interference, its value at $K = 30$ is still comparable to the reference value of 0.05 b/s/Hz that can be achieved with an LTE Release-8 single-user (SU)-MIMO system of two transmit and two receive antennas.

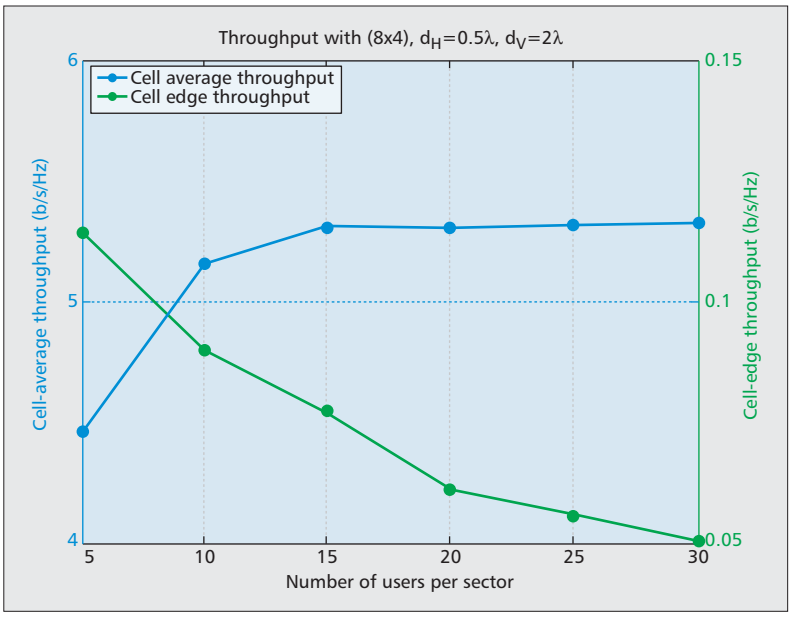


Figure 5. *8H × 4V FD-MIMO multi-user dimensioning:* $d_H = 0.5\lambda$, $d_V = 4\lambda$.

Performance Evaluation Using the Proposed 3D-SCM

This subsection shows system-level performance gains of FD-MIMO in the full-blown system-level simulation setup; that is, both long-term and short-term fading channels are modeled according to the methodology introduced earlier with elevation spread of 1.7°.

BS antenna configurations of 2H × 1V (reference), 32H × 1V with 0.5λ spacing, 8H × 4V with (0.5λ, 0.5λ) spacing, and 8H × 4V with (0.5λ, 2λ) spacing are simulated. The reference configuration of 2H × 1V is chosen as a practical configuration of the Release 10 LTE [13]. The antenna radiation pattern (3 dB beamwidth and the peak gain) is used appropriately for each configuration to ensure fair comparisons. For example, for the 2H × 1V configuration, each digital antenna port corresponds to a vertical subarray of physical antenna elements that produces horizontal and vertical 3 dB beamwidths of 70° and 10°, respectively, and a peak antenna gain of 14 dBi. For the FD-MIMO configurations (i.e., 32H × 1V and 8H × 4V), there is a one-to-one mapping between an antenna port and a physical antenna element; the antenna pattern of 90° horizontal and vertical 3 dB beamwidths for each antenna port and a peak antenna gain of 6.4 dBi is a result of a particular choice of patch antenna design. For the reference configuration, the 3GPP LTE Release 8 procedure for CSI feedback, and rank and precoder selection are used. For evaluating an upper bound of performance of the FD-MIMO configurations, ideal channel knowledge is assumed, and the all-user full-bandwidth scheduling and the MRT precoding are used as for the simulations earlier.

Our simulation results with the 3D channel model are presented in Fig. 6. Similar to what-was discussed earlier, it is observed that about 2–3.6 times cell average throughput gain and 1.5–5 times cell edge throughput gain are achieved for the FD-MIMO antenna configurations compared with the reference configuration. The spectral efficiency gain mainly comes from all-user full-bandwidth scheduling and low-interference spatial multiplexing in both the horizontal and vertical dimensions.

Channel Estimation for FD-MIMO

In order to support FD-MIMO, the scalability of reference signals and accuracy of channel estimation relying on reference signals should be carefully considered. Reference signals are necessary for demodulation of downlink data signals and for CSI estimation.

For demodulation, the Release-10 LTE specification allows up to four demodulation reference signals (DMRS) to bemultiplexed to support channel estimation of up to four UE devices' channels, each with one transmission layer. Furthermore, given that higher-order MU-MIMO is realizable by FD-MIMO, system design may consider multiplexing a larger number of DMRS (e.g., 8 or 16) for MU-MIMO, by increasing the code-division multiplexing (CDM) spreading factor or increasing the number of resource elements to carry the DMRS.

On the other hand, for CSI estimation, supporting estimation of a larger number of un-precoded channels may be necessary. However, additional overhead to achieve this should be controlled through careful design of CSI reference signals (CSI-RS). Simply configuring the full 32 CSI-RS may not incur too much overhead except for slow UE devices, in which case a longer CSI-RS time periodicity can be configured. For general UE devices, however, configuring full CSI-RS may incur prohibitively large overhead; hence, more study is needed on how to efficiently design CSI-RS for FD-MIMO.

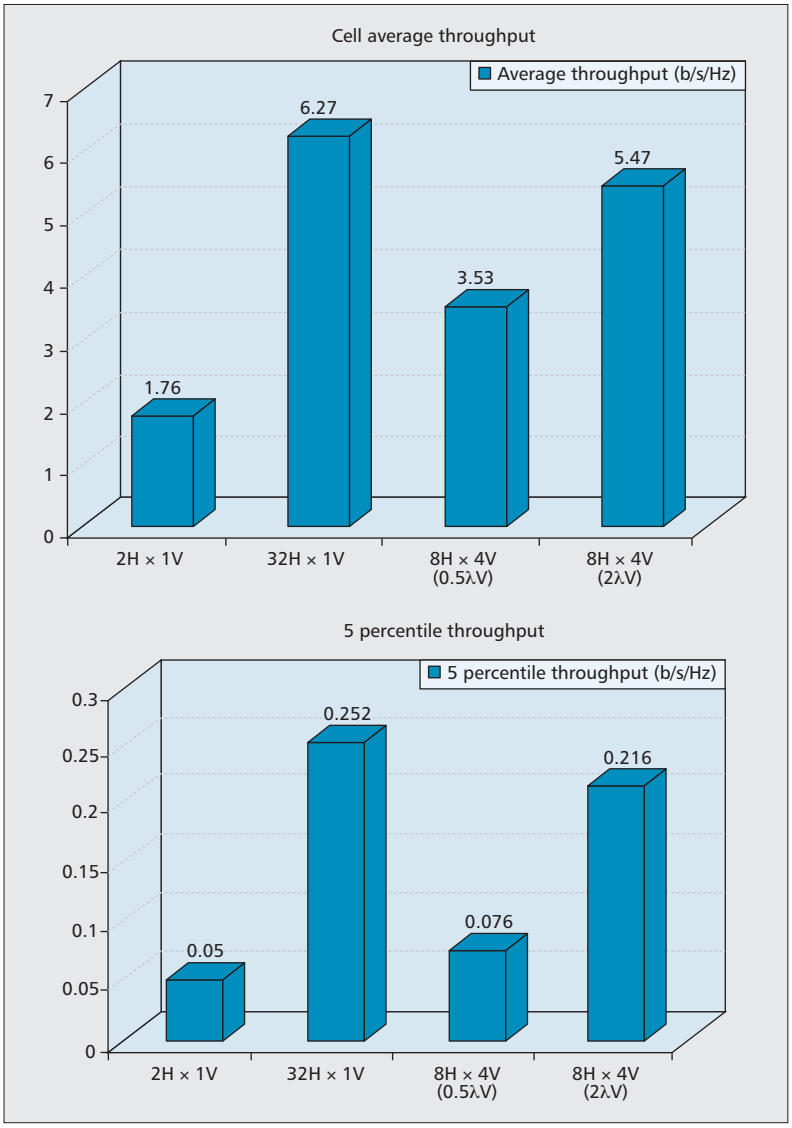


Figure 6. *Performance comparison between various 2D antenna array configurations.*

Conclusion

This article has proposed FD-MIMO, which has the potential to boost spectral efficiency by equipping and utilizing 2D active antenna arrays at base stations. To evaluate the proposed system, the 3D SCM model was introduced, which captures both azimuth and elevation angle statistics, and is able to account for both horizontal and vertical beamforming. Under the 3D SCM model, system-level performance of various base station antenna configurations has been evaluated, including 2H × 1V, 32H × 1V, and 8H × 4V. Our system-level simulation results have shown that 2–3.6 times cell average throughput gain and 1.5–5 times cell edge throughput gain can be achieved by FD-MIMO against 2H × 1V.

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BIOGRAPHIES

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